Qatar-1b

R-1442

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_201127` · created 2026-08-01T08:31:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459180.6559 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.032
Transit depth (Rp/Rs)^2	1.32 +/- 0.74 [%]
Orbital Inclination (inc)	85.45 +/- 2.11
Airmass coefficient 1 (a1)	0.936 +/- 0.018
Airmass coefficient 2 (a2)	0.0418 +/- 0.0094
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	#3 - [498, 145]
Optimal Method	PSF photometry
Transit Duration (day)	0.07 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.115 ± 0.032 vs 0.1463 (deviation 0.98σ)
Duration fit vs published	0.07 d vs 0.0692 d (deviation 0.08σ)
Mid-time O–C	3.0 min
Depth SNR	3.6
Residual scatter	1.89 %

Light curve

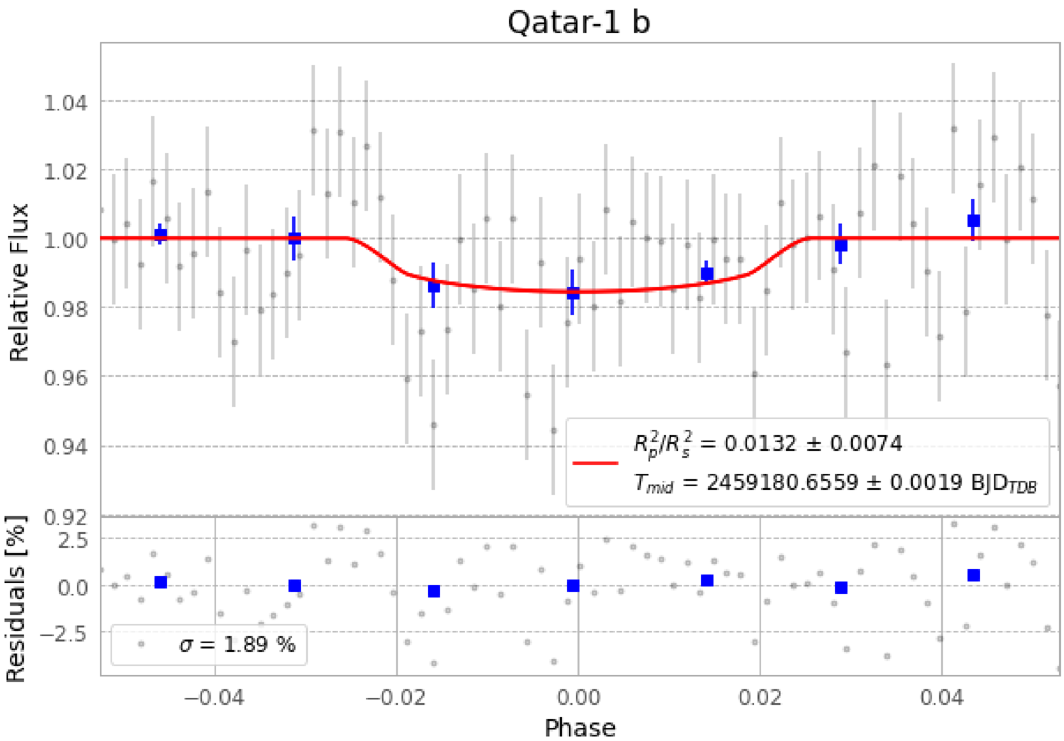

Figure 1 — FinalLightCurve_Qatar-1 b_2020-11-26.png (SHA-256 d582c52e521d464d…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [506, 338], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e8dde6c02fa1fc47…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

73 FITS frames (48 MB), `Qatar-1201127015412.FITS` → `Qatar-1201127053012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-201127.log (29b3699701b7…), FinalLightCurve_Qatar-1 b_2020-11-26.png (d582c52e521d…), FinalLightCurve_Qatar-1 b_2020-11-26.csv (8939daedfdd7…)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)